



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	AAI

Name of Student:	BARZ ANKIT VINOD
Roll No. :	61
Assignment No.:	2
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	3
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	07

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Vaishali Shinde

Signature :

Date : 17/3

Q. 1) In a certain city, 1% of the population is suspected to have a specific blood disorder. A diagnosis test is 95% effective in detecting the disease when it is present, but it also yields a false positive result for 0.5% of healthy persons. If a person is selected at random and the test result is negative, find the probability that the person actually has the disorder?

→ wkt,

$$P(\text{disease} \mid \text{Negative test})$$

S1. define the probabilities -

$$P(D) = 0.01$$

$$\& \quad P(H) = 0.99$$

Test accuracy,

$$\text{sensitivity} = 95\% \text{ means } \therefore P(\text{Positive} \mid D) = 0.95$$

$$\& \therefore P(\text{Negative} \mid D) = 0.05$$

$$\text{false positive rate} = 0.5\%$$

$$\therefore P(\text{Positive} \mid H) = 0.005$$

$$\& \therefore P(\text{Negative} \mid H) = 0.995$$

S2. Use bayes theorem -

$$P(D \mid N) = \frac{P(N \mid D) P(D)}{P(N)}$$

$$P(N) = P(N \mid D) P(D) + P(N \mid H) P(H)$$

$$= (0.05)(0.01) + (0.995)(0.99)$$

$$= 0.0005 + 0.98505$$

$$= 0.98555$$

S3.

$$P(D \mid N) = \frac{0.05 \times 0.01}{0.98555} = \frac{0.0005}{0.98555}$$

$$\approx 0.000507$$

Q. 2) In a neighbourhood 90% children were falling sick due to blue and 10% due to measles and no other diseases. The probability of observing rashes for measles is 0.95 and for blue is 0.08, if a child develops rashes, find the children's probability of having blue.

→ given,

S1-

$$P(F) = 0.90 \text{ (blue)}$$

$$P(M) = 0.10 \text{ (measles)}$$

Rashes probability,

$$P(R|M) = 0.95$$

$$P(R|F) = 0.08$$

S2- Apply Bayes theorem -

$$P(F|R) = \frac{P(R|F) P(F)}{P(R)}$$

$$\begin{aligned} P(R) &= P(R|F) P(F) + P(R|M) P(M) \\ &= (0.08)(0.90) + (0.95)(0.10) \\ &= 0.072 + 0.095 \\ &= \underline{0.167} \end{aligned}$$

S3- final calculation -

$$\begin{aligned} P(F|R) &= \frac{0.08 \times 0.90}{0.167} \\ &= \frac{0.072}{0.167} \\ &= 0.431 \end{aligned}$$

∴ 43.1%

3) Three urns are there containing white and black balls, first urn has 3 white and 2 black balls, second urn has 2 white and 3 black balls and third urn has 4 white and 1 black balls. without any biasing one urn is chosen from that one ball is chosen randomly which was white. what is probability that it came from the third urn?

→ sol.

$P(\text{Third urn} \mid \text{white ball})$

S1 let,

$U_1 = \text{first urn } (3W, 2B)$

$U_2 = \text{second urn } (2W, 3B)$

$U_3 = \text{third urn } (4W, 1B)$

Since urn is chosen without bias:

$$P(U_1) = P(U_2) = P(U_3) = \frac{1}{3}$$

$\therefore$ Probability of drawing white from each urn.

$$P(W \mid U_1) = \frac{3}{5}$$

$$P(W \mid U_2) = \frac{2}{5}$$

$$P(W \mid U_3) = \frac{4}{5}$$

32 Apply Bayes theorem,

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

$$P(U_3 \mid W) = \frac{P(W \mid U_3) P(U_3)}{P(W)}$$

33. base

$$2. P(W) = 2P(W|U)P(U)$$

$$= \frac{2}{5} \cdot \frac{1}{3} + \frac{2}{5} \cdot \frac{1}{3} + \frac{4}{5} \cdot \frac{1}{3}$$

$$P(W) = \frac{1}{3} \left(\frac{2+2+4}{5} \right)$$

$$= \frac{1}{3} \cdot \frac{8}{5}$$

$$= \frac{8}{15}$$

$$= \frac{3}{5}$$

34. base,

$$P(U_2|W) = \frac{\frac{4}{3} \cdot \frac{1}{3}}{\frac{3}{5}}$$

$$= \frac{\frac{4}{15}}{\frac{3}{5}}$$

$$= \frac{4}{15} \cdot \frac{5}{3}$$

$$= \frac{20}{45}$$

$$= \frac{4}{9}$$

$$\approx 0.4444$$